

Robin Lange

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Robotics and Autonomous Systems | Perception, Mission Logic and Physical Testing

Education

Georgia Southern University **Aug. 2023–May 2027**
B.S. Mechanical Engineering, Vehicle Design & Testing concentration (in progress) *GPA: 4.00/4.00*
Institutional GPA through Spring 2026; President's List in every completed graded term. Honors College Scholar; Tau Beta Pi; Allen E. Paulson Scholar.

Descartes-Gymnasium Neuburg an der Donau **Sept. 2014–June 2023**
Allgemeine Hochschulreife / General Higher Education Entrance Qualification *Germany*
Final German grade: 1.2 (1.0 highest); 791/900 points.

Research Experience

Institute for Human & Machine Cognition (IHMC) **May–Aug. 2026**
National Center for Collaborative Autonomy (NCCA)
Collaborative Robotics Autonomy Intern *Pensacola, FL*

- Built the Rust/Python autonomy codebase from scratch for a collaborative platform with two Boston Dynamics Spot robots; implemented pickup/drop-off sequencing and Joint Activity Graph (JAG) coordination.
- Combined inverse kinematics for coarse approach with local perception for final grasping, enabling successful hardware pickups despite tracking drift and imperfect perception.
- Implemented coordinate transforms and Zenoh communication; used Motive motion capture, MCAP/Foxglove logging and robot SDKs; added tracking markers to a manually operated Husky AGV.

Georgia Southern University **Oct. 2023–present**
Undergraduate Researcher, Intelligent Vehicles Navigation Research Laboratory *Statesboro, GA*

- Integrated visual SLAM, nvblox mapping, Nav2 and behavior-tree mission logic on Jetson/RealSense UAV hardware; evaluated localization in a 110-second cart trial (14.5 cm RMSE on valid reference segments).
- Developed a LiDAR hand-motion interface for UAV control; coauthored two published SAE papers on lane perception and vehicle-simulation calibration.
- Developing a MATLAB/AVL longitudinal-control baseline and simulator validation for an Honors thesis; learned-prediction comparisons are planned.

CARISSMA Institute of Safety in Future Mobility **July 1–Aug. 15, 2025**
Technische Hochschule Ingolstadt
Vehicle Testing Research Intern *Ingolstadt, Germany*

- Designed and built the covering, quick-assembly setup and fold-away mechanisms of a soft truck-rear crash target, developing an earlier CAD/foam concept into a testable physical prototype.
- Researched radar, LiDAR and camera sensing; supported vehicle tests and analyzed qualitative LiDAR/4D-radar returns with team scripts.

Fraunhofer Application Center **June 1–Aug. 11, 2024**
“Connected Mobility and Infrastructure” – Fraunhofer IVI
Autonomous Systems and AI Intern *Ingolstadt, Germany*

- Implemented hybrid state-machine/behavior-tree mission intelligence in Python, ROS 2 and py_trees for ALBACOPTER, translating flight-phase requirements into executable logic.
- Verified emergency interruption across flight phases, evaluated resource use and documented ALBACOPTER simulation results.

Technical Skills

Programming: Python, Rust, MATLAB, C++; functional/procedural programming; entity-component-system (ECS) patterns.

Robotics: ROS 2/DDS, Zenoh, Protocol Buffers; inverse kinematics, behavior trees, coordinate transforms, visual SLAM.

Hardware and tools: Boston Dynamics Spot, motion capture, LiDAR, RGB-D cameras, Jetson, Gazebo, SolidWorks, Git/Linux, MCAP/Foxglove.

Selected Awards and Scholarships

Third Place Undergraduate, Gulfstream Innovation, Engineering and Flight Excellence Awards – April 24, 2026. *Development of a Pilot AI for Autonomous UAV Inspection in Industry 5.0 Using Visual SLAM and Behavior Trees*; CEC Research Symposium.

Second Place Undergraduate, Georgia Power Innovation Awards – April 25, 2025. *Drone Guiding Through Equivalent Hand Movement Transversion*; CEC Research Symposium.

NCUR Engineering Division Video Contest Awardee – Council on Undergraduate Research, 2026.

Honors College Continuing Scholarship (2024–27); Charles Robert Pound, Sr. Scholarship (2026–27); Paulson Scholar (2025); Vickie Hawkins Scholarship (2024).

Published Research

Soloiu, V., Sutton, T., Mehrzed, S., **Lange, R.**, Zimmerman, C., and Peralta Lopez, G. “Data-Driven Calibration of Vehicle Simulations for Accurate Control Response in Development and Testing.” *SAE Technical Paper 2026-01-0058*, 2026. doi:10.4271/2026-01-0058.

Soloiu, V., Mehrzed, S., Kroeger, L., Pierce, K., Sutton, T., and **Lange, R.** “Development of a Rule-Based Lidar Lane-Keeping Assistance System for Autonomous Vehicles Applications.” *SAE Technical Paper 2025-01-8025*, 2025. doi:10.4271/2025-01-8025.

Preprint and Manuscript in Preparation

Soloiu, V., Sutton, T., **Lange, R.**, and Zimmerman, C. “Warehouse Platooning Framework Using a Sensor-Rich Leader and Minimal Perception Follower.” SSRN 7067358. Preprint; not peer reviewed.

Lange, R., Sezgin, F., Inderst, M., Sutton, T., Lugner, R., Brandmeier, T., and Soloiu, V. “Design and Validation of a Soft Crash Target for Multimodal Sensor-Based Truck Underride Testing.” Manuscript in preparation.

Lange, R. “When to Trust Learned Lead-Vehicle Predictions: Reliability-Gated Preview for Safety-Supervised Adaptive Cruise Control.” Honors thesis in progress; journal paper planned.

Additional Research Presentations

Southern Regional Honors Council Conference, 2026: Intelligent UAV Navigation with V-SLAM and Behavior Trees for Industry 5.0 Applications.

Southern Regional Honors Council Conference, March 2025, Chattanooga, TN: 3D-Space Hand Trajectory Tracking for Intuitive UAV Control.

GS4 symposium: Enhancing UAV Command through 3D-LiDAR-Driven Hand Movement Recognition.

Teaching and Academic Preparation

Physics I/II teaching assistant (Jan. 2025–present): evaluate experimental reports. **Computing for Engineers laboratory assistant** (Jan.–May 2024): guided MATLAB and programming labs. **First-Year Experience peer leader** (Aug.–Dec. 2024).

Completed coursework: Calculus I–III, Differential Equations, Numerical Methods, Dynamics, Mechatronics, Circuit Analysis, Program Design, Logic Circuit Design.

Languages: German (native), English (fluent), French (DELF B1).